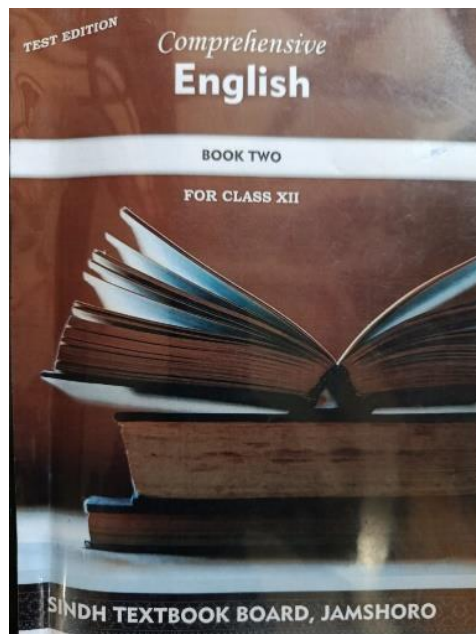


English
Book Two
For the class XII



Climate Change



Short Question-Answers

Q-1 How does climate change affect water availability in different regions?

Ans- Climate change significantly affects water availability across different regions by altering rainfall patterns and increasing global temperatures. In many areas, rising temperatures accelerate evaporation, which reduces water levels in rivers, lakes, and soil. As a result, several regions experience severe droughts and growing water scarcity. Climate change also intensifies agricultural and ecological droughts, harming crops and ecosystems. In addition, expanding deserts reduce fertile land for food production. Dust and sandstorms may also increase.

Consequently, millions of people face the growing threat of insufficient water for daily life and agriculture.

Q-2 How does climate change impact biodiversity and species survival?

Ans- Climate change significantly threatens biodiversity and the survival of many species. Rising global temperatures, extreme weather events, forest fires, and changing habitats make it difficult for plants and animals to survive in their natural environments. Many species are forced to migrate to cooler or more suitable regions, but not all can adapt or relocate successfully. Climate change also increases the spread of pests and diseases, further endangering wildlife. As a result, species are disappearing at an alarming rate, with about one million species at risk of extinction in the coming decades.

Q-3 How is the increase in poverty an effect of the climate change?

Ans- Climate change contributes to the increase in poverty by damaging the natural and economic resources on which many people depend for survival. Extreme weather events such as floods, droughts, and storms destroy homes, crops, and livestock, reducing people's income and food supply. Farmers and fishermen often suffer heavy losses when climate conditions become unpredictable. In addition, rising sea levels and environmental disasters force many communities to relocate, leading to unemployment and economic instability. As a result, vulnerable populations, especially in developing countries, face greater hardship, deepening poverty and reducing opportunities for a stable livelihood.

Q-4 What is climate change and what are its main causes?

Ans- Climate change refers to long-term changes in the Earth's average weather patterns, including temperature, rainfall, wind, and other climatic conditions. It occurs due to both natural processes and human activities, but human actions have become the major driving force in recent decades. The main causes include the burning of fossil fuels such as coal, oil, and natural gas, which release large amounts of carbon dioxide into the atmosphere. Other significant causes are industrial activities, deforestation, agricultural practices that emit methane and nitrous oxide, transportation using fossil fuels, poor waste management, and increasing urbanization and overconsumption.

Q. 5 How does deforestation contribute to climate change?

Ans- Deforestation contributes to climate change by reducing the number of trees that absorb carbon dioxide from the atmosphere. Trees naturally store carbon, but when forests are cut down or burned, the stored carbon is released back into the air as carbon dioxide, a major greenhouse gas. This increases the concentration of greenhouse gases in the atmosphere, which intensifies global warming. In addition, destroying forests weakens nature's ability to regulate the climate. Large-scale deforestation for farming, grazing, and development therefore significantly accelerates climate change and environmental imbalance.

Q. 6 Why might people in small island nations be more affected by climate change?

Ans- People living in small island nations are more affected by climate change because they are highly vulnerable to rising sea levels and extreme weather events. As global temperatures increase, melting ice and warming oceans cause sea levels to rise, which can flood coastal areas and

contaminate freshwater sources with saltwater. This threatens homes, agriculture, and drinking water supplies. In some cases, entire communities may be forced to relocate. Additionally, stronger storms and cyclones can cause severe damage to infrastructure and livelihoods, making it difficult for these nations to recover and adapt.

Q. 7 Why is the ocean both a victim and a helper in the fight against climate change?

Ans- The ocean is both a victim and a helper in the fight against climate change. It acts as a helper by absorbing a large portion of the heat and carbon dioxide produced by global warming, which helps reduce the amount of heat and greenhouse gases in the atmosphere. However, this process also harms the ocean itself. As it absorbs more heat, ocean temperatures rise and water expands, causing sea levels to increase. Moreover, absorbing excess carbon dioxide makes the ocean more acidic, which damages marine life, coral reefs, and aquatic ecosystems.

Q.8 Do you think switching renewable energy can solve all climate problems? Why or why not?

Ans- Switching to renewable energy can greatly help reduce climate change, but it cannot solve all climate problems by itself. Renewable sources such as solar, wind, and hydro power reduce the use of fossil fuels and lower greenhouse gas emissions. However, climate change is also caused by other factors like deforestation, industrial activities, agricultural practices, waste management, and overconsumption. Therefore, along with renewable energy, people must protect forests, adopt sustainable farming, reduce waste, and promote responsible consumption. Only a combination of these actions can effectively address climate change and protect the planet.

Q.9 What connection can you draw between climate change and poverty?

Ans- Climate change and poverty are closely connected because environmental changes often damage the resources that poor communities depend on for survival. Extreme weather events such as floods, droughts, and storms destroy crops, homes, and livestock, reducing people's income and food supply. Many poor families rely on agriculture, fishing, or natural resources, which are highly affected by changing climate conditions. In addition, disasters can force people to migrate and create unemployment and economic instability. As a result, climate change deepens poverty and makes it harder for vulnerable populations to improve their living conditions.

Q.10 Why is climate change considered a global issue, not just a local or international one?

Ans- Climate change is considered a global issue because its causes and effects extend across the entire planet and cannot be confined to one country or region. Greenhouse gases released in one part of the world spread through the atmosphere and affect the global climate system. As a result, rising temperatures, sea-level rise, extreme weather events, and ecosystem disruptions impact people everywhere. No nation can fully escape its consequences. Therefore, addressing climate change requires collective international cooperation, shared responsibility, and coordinated action to protect the Earth and ensure a sustainable future for all.

Below are clear meanings and common synonyms for each word / phrase, written in simple, exam-friendly language:

1. **Account for**

Meaning: To explain the cause or reason for something.

Synonyms: explain, justify, clarify.

2. **Adaptation**

Meaning: The process of adjusting to new conditions or environment.

Synonyms: adjustment, modification, accommodation.

3. **Alternation**

Meaning: The act of occurring in turns or repeatedly changing between two states.

Synonyms: rotation, fluctuation, succession.

4. **Anaerobic**

Meaning: Occurring without oxygen.

Synonyms: oxygen-free, airless.

5. **Arctic**

Meaning: The extremely cold region around the North Pole.

Synonyms: polar, frigid, icy.

6. **Biodiversity**

Meaning: The variety of plant and animal life in a particular environment.

Synonyms: biological diversity, variety of life.

7. **Combat**

Meaning: To fight against something.

Synonyms: fight, oppose, battle.

8. **Combustion**

Meaning: The process of burning.

Synonyms: burning, ignition.

9. **Composting**

Meaning: The process of turning organic waste into natural fertilizer.

Synonyms: organic recycling, decomposition process.

10. **Conscious**

Meaning: Aware of something; having knowledge or awareness.

Synonyms: aware, mindful, alert.

11. Conserve

Meaning: To protect and preserve resources.

Synonyms: preserve, protect, save.

12. Coral Reef

Meaning: A ridge of coral formed in warm ocean waters that supports marine life.

Synonyms: coral formation, marine reef.

13. Decomposition

Meaning: The process of breaking down organic matter.

Synonyms: decay, rotting, disintegration.

14. Ecosystem

Meaning: A community of living organisms interacting with their environment.

Synonyms: ecological system, biological community.

15. Emission

Meaning: The release of gases or substances into the air.

Synonyms: discharge, release.

16. Enforce

Meaning: To make sure a law or rule is obeyed.

Synonyms: implement, impose, apply.

17. Escalate

Meaning: To increase or become more intense.

Synonyms: intensify, increase, heighten.

18. Exacerbate

Meaning: To make a problem or situation worse.

Synonyms: aggravate, worsen, intensify.

19. Extinct

Meaning: No longer existing; completely died out.

Synonyms: vanished, wiped out.

20. Incineration

Meaning: The process of burning waste materials.

Synonyms: burning, cremation.

21. Intrusion

Meaning: An unwanted entry or interference.

Synonyms: invasion, interference.

22. Invasive

Meaning: Spreading harmfully into a new area.

Synonyms: encroaching, aggressive.

23. Landfills

Meaning: Places where waste is buried in the ground.

Synonyms: dumping grounds, waste disposal sites.

24. Manure

Meaning: Animal waste used as fertilizer for crops.

Synonyms: dung, organic fertilizer.

25. Mitigation

Meaning: The act of reducing the severity of something.

Synonyms: reduction, alleviation.

26. Pollutants

Meaning: Harmful substances that contaminate the environment.

Synonyms: contaminants, toxins.

27. Potent

Meaning: Very powerful or effective.

Synonyms: strong, powerful.

28. Precipitation

Meaning: Water falling from the atmosphere such as rain or snow.

Synonyms: rainfall, snowfall.

29. Protracted

Meaning: Lasting for a long time.

Synonyms: prolonged, extended.

30. Soak up

Meaning: To absorb liquid or substance.

Synonyms: absorb, take in.

31. Stir

Meaning: To move or cause movement; to provoke or arouse.

Synonyms: mix, agitate, provoke.

32. Sustainable

Meaning: Able to continue for a long time without harming the environment.

Synonyms: maintainable, eco-friendly.

33. Tropical

Meaning: Relating to the hot regions near the equator.

Synonyms: equatorial, hot-climate.

34. Urbanization

Meaning: The growth and expansion of cities.

Synonyms: city development, metropolitan expansion.

35. Variables

Meaning: Factors that can change or vary.

Synonyms: factors, elements.

36. Vulnerable

Meaning: Easily harmed or at risk.

Synonyms: weak, exposed, susceptible.

15 MCQs based on the essay “Climate Change”:

1. What does climate change refer to?
 - A) Short-term fluctuations in local weather
 - B) Long-term changes in Earth’s average weather patterns
 - C) Natural disasters only
 - D) Seasonal rainfall variations
2. Which human activity is the largest contributor to greenhouse gas emissions?
 - A) Urban gardening
 - B) Fossil fuel burning
 - C) Hiking and tourism
 - D) Traditional fishing
3. How does deforestation increase climate change?
 - A) By creating more shade
 - B) By reducing CO₂ absorption and releasing stored carbon
 - C) By increasing rainfall
 - D) By reducing soil fertility only
4. Which greenhouse gas is primarily produced by livestock farming?
 - A) Carbon dioxide
 - B) Methane
 - C) Nitrous oxide
 - D) Ozone

5. Why are small island nations particularly vulnerable to climate change?
 - A) They have fewer natural resources
 - B) They face sea-level rise and flooding
 - C) They have advanced technology
 - D) They are mostly industrialized
6. How does industrial activity contribute to climate change?
 - A) By reducing the number of forests
 - B) By emitting greenhouse gases like CO₂, CH₄, and N₂O
 - C) By increasing rainfall
 - D) By creating deserts
7. Which of the following is a major effect of climate change on oceans?
 - A) Decreased acidity
 - B) Sea-level rise and ocean warming
 - C) Formation of freshwater lakes
 - D) Reduced salinity
8. How does transportation contribute to climate change?
 - A) By promoting cycling and walking
 - B) By emitting greenhouse gases from fossil fuels
 - C) By increasing rainfall
 - D) By improving air quality
9. Which of the following worsens food scarcity due to climate change?
 - A) Fertile soil
 - B) Increased rainfall
 - C) Crop failure and declining livestock productivity
 - D) Forest protection
10. How does climate change impact human health?
 - A) Reduces exposure to diseases
 - B) Causes heat stress, malnutrition, and disease spread
 - C) Improves sanitation
 - D) Increases immunity globally
11. Which of these contributes to rising temperatures globally?
 - A) Reforestation
 - B) Greenhouse gas emissions
 - C) Use of renewable energy
 - D) Efficient recycling
12. What role does the ocean play in climate change?
 - A) Absorbs heat and CO₂, but suffers warming and acidification

- B) Reduces the greenhouse effect permanently
 - C) Prevents droughts completely
 - D) Produces fossil fuels naturally
13. Why is climate change considered a global issue?
- A) Because only industrial countries are affected
 - B) Because its causes and effects affect the entire planet
 - C) Because it occurs only during summer
 - D) Because only farmers notice it
14. Which of the following human activities produces methane in waste management?
- A) Recycling plastics
 - B) Decomposition of organic waste in landfills
 - C) Solar energy production
 - D) Using electric vehicles
15. What is necessary to effectively combat climate change?
- A) Individual, community, and global actions
 - B) Relying only on renewable energy
 - C) Ignoring local adaptation strategies
 - D) Focusing solely on industrial growth

Answer Key & Justifications

1- **B) Long-term changes in Earth's average weather patterns**

Justification: Climate change refers to sustained alterations in average temperature, rainfall, wind, and other climatic conditions, not short-term or seasonal variations.

2- **B) Fossil fuel burning**

Justification: Coal, oil, and natural gas release large amounts of CO₂ when used for electricity, heat, and transportation, making fossil fuels the largest human contributor to greenhouse gas emissions.

3- **B) By reducing CO₂ absorption and releasing stored carbon**

Justification: Trees absorb CO₂; when cut, they release stored carbon, increasing greenhouse gases in the atmosphere.

4- **B) Methane**

Justification: Livestock farming, especially cows and sheep, produces methane during digestion, a potent greenhouse gas.

5- **B) They face sea-level rise and flooding**

Justification: Small island nations are highly vulnerable because rising oceans threaten homes, freshwater, and livelihoods.

6- **B) By emitting greenhouse gases like CO₂, CH₄, and N₂O**

Justification: Industrial processes burn fossil fuels and release gases such as carbon dioxide, methane, and nitrous oxide, contributing to global warming.

7- **B) Sea-level rise and ocean warming**

Justification: Oceans absorb heat and expand; melting ice adds water, causing sea-level rise, and absorbed CO₂ increases acidification, affecting marine life.

8- **B) By emitting greenhouse gases from fossil fuels**

Justification: Cars, trucks, planes, and ships predominantly run on petroleum products, releasing CO₂ and other greenhouse gases.

9- **C) Crop failure and declining livestock productivity**

Justification: Droughts, heat stress, and changing climate disrupt food production, leading to scarcity and hunger.

10- **B) Causes heat stress, malnutrition, and disease spread**

Justification: Climate change increases extreme temperatures, spreads diseases, affects food security, and stresses healthcare systems.

11- **B) Greenhouse gas emissions**

Justification: Rising concentrations of CO₂, CH₄, and N₂O trap heat in the atmosphere, increasing global temperatures.

12- **A) Absorbs heat and CO₂, but suffers warming and acidification**

Justification: Oceans help reduce atmospheric heat and carbon but warm and acidify in the process, harming marine life.

13- **B) Because its causes and effects affect the entire planet**

Justification: Greenhouse gases and warming affect global climate systems, making climate change a worldwide issue.

14- **B) Decomposition of organic waste in landfills**

Justification: Methane is released when organic waste decomposes anaerobically in landfills.

15- **A) Individual, community, and global actions**

Justification: Effective climate action requires combined efforts at all levels—individual behavior, community projects, and global policies—to mitigate and adapt to climate change.

Essay

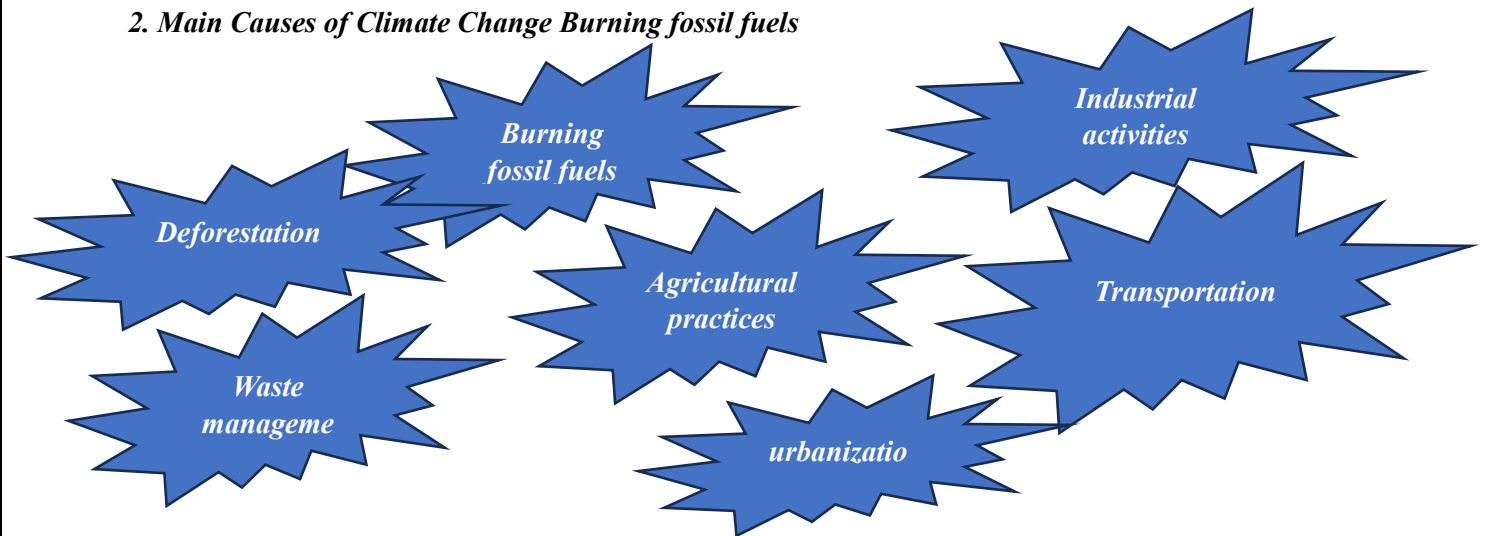
Brainstorming Points on Climate Change

1. Introduction

- Long-term changes in Earth's weather patterns (temperature, rainfall, wind, etc.)

- *Caused by natural processes and human activities*

2. Main Causes of Climate Change Burning fossil fuels

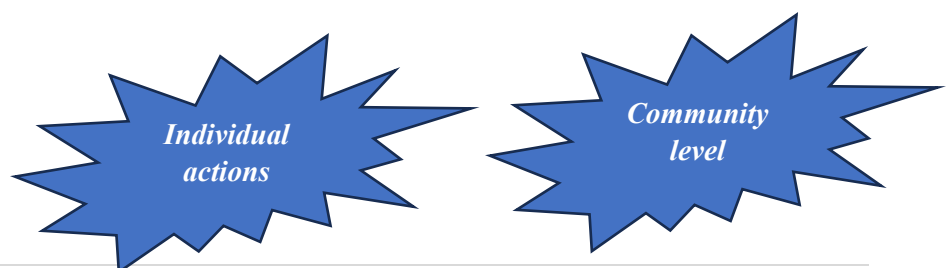


3. Major Effects / Impacts



4. Vulnerable Populations

5. Solutions / Way Forward



6. Conclusion

Climate Change: Causes, Effects, and the Way Forward

Climate change refers to long-term alterations in the Earth's weather patterns, including temperature, rainfall, wind, and other climatic conditions, caused by both natural processes and human activities. Human activities, especially the emission of greenhouse gases from burning fossil fuels, industrial processes, deforestation, and unsustainable agricultural practices, have accelerated global warming. Transportation, waste mismanagement, overconsumption, and urbanization further intensify the problem. Together, these factors increase greenhouse gas concentrations in the atmosphere, leading to significant environmental and social consequences.

The effects of climate change are widespread and alarming. Rising global temperatures result in more heatwaves and make outdoor work and daily life challenging, while extreme weather events, including storms, cyclones, and hurricanes, cause destruction and economic losses. Water availability is severely affected, with droughts reducing water supply for drinking, agriculture, and ecosystems, and expanding deserts threatening food production. Oceans, which absorb heat and carbon dioxide, are warming and becoming more acidic, endangering marine life and raising sea levels. Wildlife and biodiversity are also at risk, with species facing extinction due to habitat loss, temperature changes, and invasive pests and diseases. Climate change further increases health risks, food scarcity, and poverty, disproportionately affecting vulnerable populations in developing countries and small island nations.

Addressing climate change requires urgent and collective action at individual, community, and global levels. People must adopt sustainable lifestyles by conserving energy, reducing waste, using public transport, and making climate-conscious choices in food and consumption. Communities should promote environmental awareness, invest in green infrastructure, and support local conservation projects. Globally, governments must implement policies to reduce emissions, transition to clean energy, and provide support for climate adaptation and mitigation.

Climate change is a growing global crisis with far-reaching effects on the environment, human health, food security, and economic stability. Only through coordinated, sustained action and shared responsibility can humanity slow its progress and secure a sustainable and livable planet for future generations.

